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Repairing Corrosion Resistant Alloy Clad Pipelines: The Evolution of Sealing Technology

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Abstract

This paper aims to compare the different sealing methods that are typically used in the subsea mechanical connectors, such as elastomeric, graphite and the frontal metal-to-metal sealing system implemented in the novel subsea connectors purposely developed for CRA clad and lined pipelines. The frontal metallic sealing has been conceived to cope with the inner layer with the same material of the CRA, thus avoiding material heterogeneity and perfectly reinstating the original containment.

Repair connectors for CRA clad and lined pipelines follow the main installation steps, diver or diverless, of the more conventional elastomeric connector and of the graphite seal connector, which rely on the pipe's outer surface for sealing. However, the peculiarity of the CRA pipeline layering calls for a new sealing method to the pipe frontal cut end to reinstate the original fluid containment and prevent to be in contact with the pipe carbon steel wall. This needs different pipe preparation to remove all flaws up to required tolerances from the pipe cut end frontal surface.

The different seal materials can be seen as an evolution to respond to the increasing repair needs:

- The **elastomeric seal** represents the largely used solution for non-sour service pipeline, the sealing is on the pipe's outer surface and requires the removal of the pipe coating and the weld seam, thanks to the polymer adaptability once energized. The repair with elastomeric seal is regarded as a non-permanent solution due to the aging of the polymer, potentially no compliant with the residual pipeline lifetime.
- The **graphite seal** is considered an evolution of elastomeric seal, replaced by metallic material making the repair permanent. Likewise for the elastomeric connector the sealing is on the outer surface, however the installation tolerances are more stringent due to the limited capability of the graphite to radially expand when energized, making the intervention more complex.
- The **frontal metal sealing**, conceived for the CRA pipelines, mates the inner CRA layer with the same noble material preventing the corrosive fluid to contact the carbon steel pipe wall, therefore reinstating the original containment differently from the outer surface sealing. The installation requires preparation of the pipe cut-end surface to the required tolerances with limited complexity compared to the elastomeric connector.

The Saipem Grip & Metal Seal Connector (GMSC), specifically designed for application to sour-service pipelines and in particular to CRA clad or lined lines, to overcome the limitations of the conventional connectors that rely on sealing on the pipe's outer surface, thus exposing the pipe wall to corrosive fluid. The frontal metal seal in fact reinstates the original pipe containment without narrowing down the internal diameter keeping the same pigging capability.

Introduction

Subsea pipeline connections utilizing diver-installed bolted flanges have been in use at least since the 1940s. By the early 1970s, the emergence of dedicated subsea engineering enabled more advanced mechanical connectors to be employed for repairing damaged or corroded pipelines.

This paper explores the evolution of non-welded mechanical connectors, primarily used in subsea pipeline repairs involving sectional replacement while the pipeline remains on the seabed. The core function of these connectors is to reinstate both pressure containment and structural integrity of the pipeline. A repair is considered permanent if its design life meets or exceeds the remaining life of the original pipeline; otherwise, it is classified as temporary.

The advantages of a seabed repair using mechanical connectors over a surface or hyperbaric welding method are well documented and include minimizing disturbance to the pipeline on the seabed, relatively quick intervention if spares are held in stock, no requirement for a pipelay vessel or specialized hyperbaric spread. However, such advantages are realized only if the resultant sealing and structural integrity, reliability and remaining life expectancy are comparable to that of the original pipeline.

Since their inception, mechanical connectors have evolved alongside pipeline technology, which has seen increases in diameter, operating pressures and temperatures, and the handling of more corrosive fluids—from crude oil to acid and sour wet gas. Pipe materials have also advanced, with improvements in yield strength, corrosion resistance, and manufacturing techniques such as long-seam welding and seamless construction. Connector designs have generally kept pace with these developments.

In the current energy environment, driven by growing demand for natural gas, the deployment of high-pressure, large-diameter sour gas trunklines with corrosion-resistant alloy (CRA) cladding or lining is becoming more common. This paper examines the latest technological advancements in mechanical connector design aimed at meeting the most onerous technical conditions posed by these challenging pipeline conditions.

Overview of Mechanical Repair Solutions

Technology development in the field of subsea repair is constantly evolving for the benefit of the energy industry, subsea repair connectors were conceived to overcome the criticalities due to the repair by welding such as water depth and pipeline conditions (age and route constraints). In this regard they represent the most viable solution for the pipeline repair



Figure 1—Repair connectors: diverless coupling (left, CSS), diver end connector (center, Oil States), diverless end connector (right, Saipem)

Subsea pipeline repair connectors are commonly used to replace the damaged sections of a pipeline reinstating the pipeline's full mechanical and pressure capacity, in those cases where damage cannot be locally repaired by other simpler means such as repair clamps (e.g., pinholes or dents). Installation sequence of a pipe repair end connector can be summarized in the following main steps whatever the sealing method or technology is implemented:

- Pipe damaged section removal
- Pipe end preparation (coating removal and cut end machining)
- End connector insertion
- End connector gripping and sealing energizing
- Installation verification (back-seal test)
- Spool connection

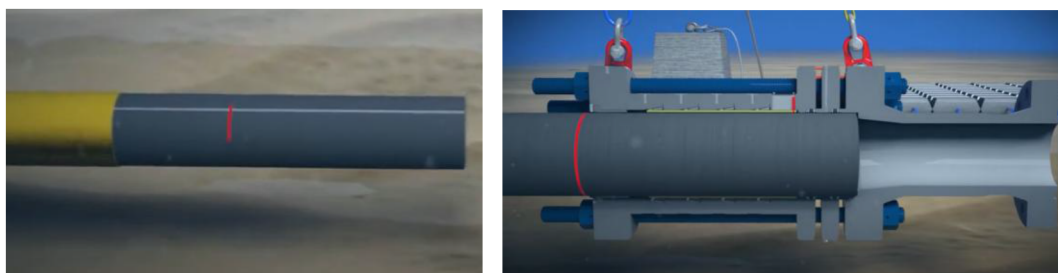


Figure 2—Pipe preparation (left) and End Connector insertion (right, CSS)

The type of connector determines the required installation equipment, including preparation tools (pipe cutting tool, pipe end/surface preparation tool), connector installation tools and pipe/spool handling tools, (such as lifting frames), used to lift the pipeline ends off the seabed and align the connectors with the spool. In the case of end-connectors, these are slid onto each pipe end and the flange make-up operation is performed by divers (if in shallow waters) or by means of special tools for diverless operations.

Connectors are provided with a gripping system to the pipeline and a sealing system. The gripping system acts on the outside surface of the pipe, with ball-cage design or wedges with indentation teeth, actuated by some sort of mechanism or by deforming the pipe against the inside of the connector which in this case has a specific profile with teeth. The sealing system is generally acting on the external surface of the pipeline, which needs to be prepared in advance to achieve the finishing conditions required by the seals, (e.g. removal of any concrete and anti-corrosion coating to bare steel, removal of pipe's longitudinal weld, etc.).

For the emergency repair of a corrosive or sour service pipeline, typically CRA pipeline or carbon steel pipeline with corrosion inhibition systems and / or a corrosion allowance, the conventional sealing systems (acting on the pipeline's external surface) would not be suitable for a permanent repair since they would allow the pipeline's external surface, in carbon steel, to be exposed to bulk fluid. Furthermore, the volume between the seal and the pipeline OD would be exposed to the transported fluid but the flow conditions in such restricted volume will be completely different from the bulk transported fluid. This volume, "dead leg" as defined by International Standards, will be exposed to the same internal pressure but in stagnant conditions and with a fluid composition that can be considerably different from the bulk transported fluid. Moreover, in such conditions it is likely that any corrosion inhibitor will not properly wet the pipe wall and will not properly provide corrosion mitigation, leaving the connector to be exposed to unpredictable underseal / crevice corrosion phenomena.

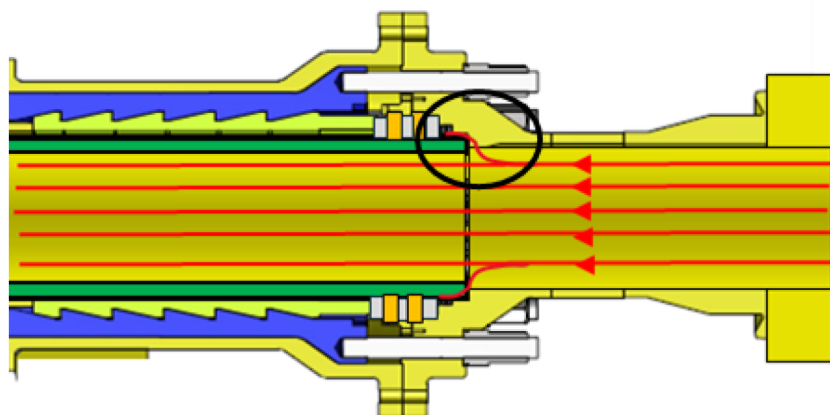


Figure 3—fluid that can penetrate in the gap between the pipe's cut end and the connector (circled in black) for external surface sealing

Therefore, the long-term sealing capability and integrity of the repair would be compromised. Instead, by sealing against the pipeline cut face, the transported bulk fluid do not come into contact with the pipeline external surface, in carbon steel thus avoiding the risk of above-mentioned uncontrollable corrosion phenomena. Using opportunely engineered metallic seals, the original material continuity of the ID of the pipeline can be reinstated, either the CRA inner layer or the carbon steel. This aspect represents the technological challenge for this specific type of service.

Mapping Solutions: Elastomers vs Graphite vs Metal

The primary material families typically available for the Pipeline Repair Connectors sealing function are the following: elastomeric, graphite and metallic.

The first connectors in the market and those that have been largely used, have been the ones that rely on the elastomeric gasket sealing on the outer surface of the pipe. The surface is machined in order to remove the external coating (concrete weight coating (CWC), thermal insulation coating and anticorrosion layer) to the bare pipe, where present the longitudinal weld seam is removed as well, the final surface finishing is according to the roughness required to guarantee the perfect matching once the seal is energized avoiding any leakage over the service life. The use of elastomeric has been largely use for decades thanks to robustness of the solution, the easy installation procedure and cost-effective repair component and associated intervention spread whether with diver or diverless. It starts facing the first criticalities when the increasing use of sour service pipeline became significant in the industry and operators starts asking for a new and reliable solution to respond to this new need, elastomeric seal indeed present a strong limitation in term of durability, it's not recognized as a permanent repair method due to the aging and degradation of the polymeric, and it's not suitable for being in contact with corrosive fluid. In this regards elastomeric seal remains an important solution in the midstream sector where the typical transportation is processed fluid or dry sweet gas, while for the upstream sector, as said, the increasing presence of sour fluid is limiting its use.

Compared to elastomeric, the graphite seal represents a first technology evolution to respond to the lifetime of the repair, the graphite does not degrade over time, meaning that the repair can be considered permanent or at least not less than the pipeline remaining service life, an additional important consideration is that it is suitable even for long-term storage without planning for any midterm replacement. The graphite seal thanks to its chemical stability is fully compliant with NACE requirements, and this represented the first important characteristic for this material to be used in sour service conditions, however the sealing method that relies on the external pipe surface poses some problems regarding the complete suitability of

this type of connector for the repair with corrosive fluid transported, because does not prevent the media to be in contact with mother pipe steel, this is the specific case for CRA pipe.

Frontal metallic seals represent the latest evolution step to specifically respond to the new need are RTJ type-like seals, i.e. face seals that are acting on the cut surface of the pipeline end. Metallic Seals shall be engineered as metal-to-metal contact interfaces, specifically designed to match the material properties of the pipeline ID. These seals are extensively utilized in flanged connections of subsea systems, including wellhead assemblies, Xmas trees, risers, and manifolds. Materials are typically selected to match the chemical resistance to aggressive fluids as well as for their performance at high temperatures and pressures. Alloys such as Inconel Alloys are employed or Stainless Steels, including grades such as AISI 304 or 316. The sealing mechanism is primarily based on precision-machined surfaces and the elastic deformation of the metallic elements, precise alignment and energization, which together ensure a high-integrity, leak-tight seal. In subsea connectors for pipeline repair applications, metallic seals shall be qualified to prove sealing effectiveness against the surface of the cut pipe end.

ELASTOMERIC SEAL

Elastomeric seals are commonly made from materials such as Hydrogenated Nitril Butadiene Rubber (HNBR) or Fluor-elastomers (FKM). Elastomeric seals are recognized for their flexibility and ease of installation, offering good chemical resistance to transported fluids in oil and gas pipeline systems, even when exposed to fuels and relatively high temperatures. Different compounding can be made to tailor the elastomer properties to the specific application increasing for example material strength, compatibility to aggressive fluid and adapting the range of applicable temperatures. Various types of elastomer compounds can be specifically formulated to tailor the material's properties to each application. This process can enhance attributes such as mechanical strength, compatibility with aggressive fluids, and the operational temperature range. The primary limitations of sealing systems that utilize elastomeric gaskets pertain to their performance under high-pressure and high-temperature conditions. In such demanding applications, alternative sealing materials may be required to ensure reliable leak prevention in pipeline systems. As a general rule, the temperature limitation for this type of seal is approximately 60°C. However, products designed for higher temperatures are available, with temperature resistance qualified up to 120°C for sweet service and limited to 80°C for sour service, as defined by NORSOK M-710. Compatibility with fluid, particularly in the presence of special additives, is to be evaluated and qualified on a case-by-case basis.

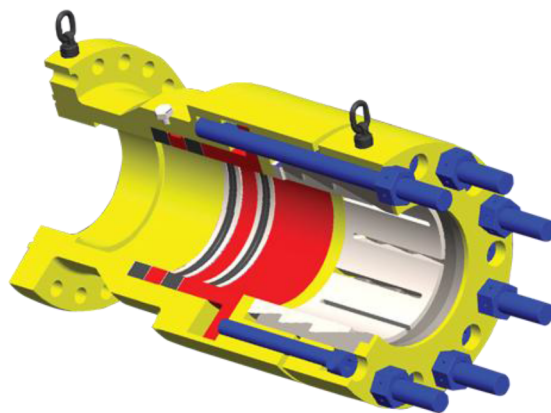


Figure 4—Typical grip and seal package for a subsea repair connector (Picture credits: Titan Subsea Innovations)

GRAPHITE SEAL

Graphite seals are valued for their exceptional thermal resistance and chemical stability. These seals demonstrate robust performance in harsh environments and under elevated pressures, retaining their sealing capability even when subjected to thermal cycling and mechanical vibration, providing exceptionally high

performance and long life. In conditions where there are thermal fluctuations the graphite element presents similar coefficient of thermal expansion to the pipeline and therefore expand and contract with the pipeline. In contrast, graphite seals require precise installation tolerances to minimize the sealing gap between the unset seal and the pipeline's outer surface. Consequently, these seals typically necessitate larger volumes and increased weight, which is especially significant in applications involving pipelines of greater diameters.

From an installation perspective the connector with graphite sealing presents many similarities with the elastomeric sealing connector, relying on the external surface of the pipe, the energizing method is like the elastomeric where the longitudinal loads from tightening system, force a metal-graphite-metal composite seal to squeeze radially against the surface. The two retaining metal rings distribute the force circumferentially the graphite seal, the mechanical properties of graphite are suitable for long terms durability under loads in an elastic condition to perfectly respond to the pipe loads and deformation changes.

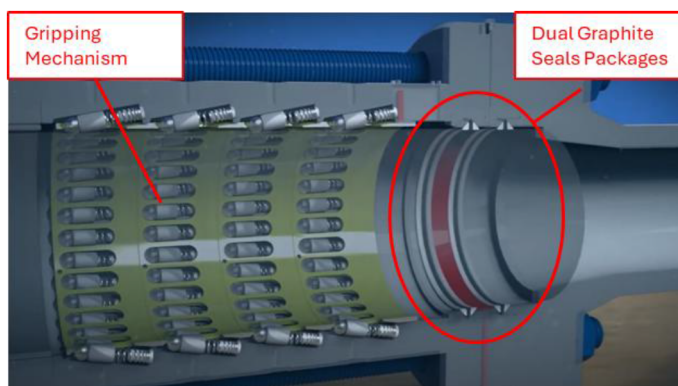


Figure 5—CSS Grip and Seal Connector with Graphite Seals Packages (Picture credits: CSS)

The selection of the metallic material of the sealing package is such as preventing any galvanic corrosion effect with the pipeline avoiding material heterogeneity.

The peculiarities of the graphite seal makes the installation demanding, the limited radial expansion reduces the installation tolerances of the connector, requiring sometime important equipment to install it within the required precision, especially when the connector is for large diameters and in a remote installation mode.



Figure 6—CSS Connector (left) and Equinor Installation Equipment (right) -(photos credits CSS)

FRONTAL METAL SEAL

The reliable production and transportation of energy supplies have necessitated more complex technological solutions for pipelines, transitioning from conventional carbon steel to CRA clad pipelines. This transition has created the need to develop new dedicated technologies for the repair of damaged pipelines,

i.e. permanently reinstating the internal CRA continuity, exemplified by the "first-of-its-kind" frontal metal seal connector. This innovative connector ensures the continuity of internal cladding without reducing the pipeline's internal diameter (ID), making it particularly advantageous for sour service and cladded pipelines in emergency scenarios. The development of this connector involved several key phases, including a review of existing connector technology limitations, the identification of functional requirements, and a comprehensive technology qualification testing program, developed with the support of DNV as the certification body. The connector's key features include DNV technology qualification for pipelines ranging from 6" to 38", metal-to-metal sealing technology for permanent repair, and the capability to restore the original operational functionality of the pipeline without reducing its internal diameter. Additionally, the connector can compensate for misalignments of the pipe cut-end and is versatile enough for both diver-assisted and diverless setups in deep water applications.

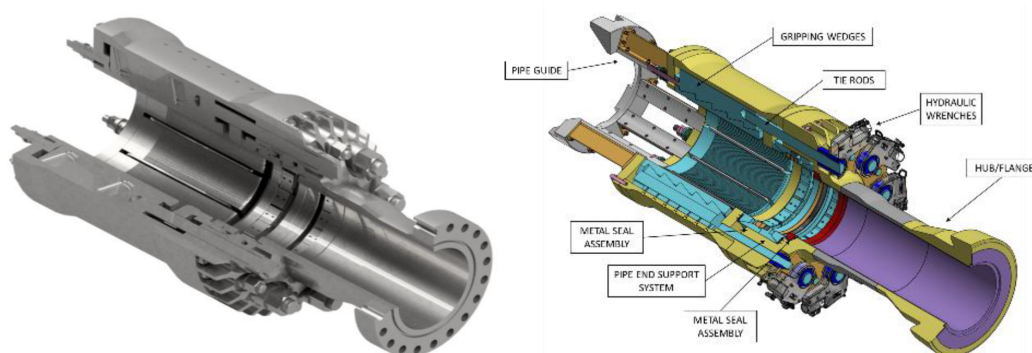


Figure 7—Saipem Grip and Metal Seal Connector (GMSC) for CRA pipeline

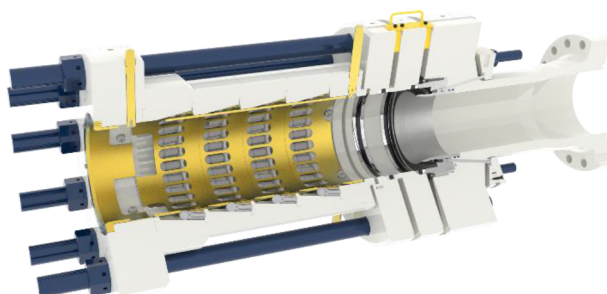


Figure 8—MORGRIP® Clad & Lined Pipe (CLiP) Connector (Picture credits: CSS)

Focus on Technology Qualification for Metal Seal

The Recommended Practice (RP) DNV-RP-F113 – Pipeline Subsea Repair establishes comprehensive criteria and guidelines for the qualification of fittings and systems employed in the repair, modification, and tie-in of subsea pipelines. This document encompasses key aspects related to the design, manufacturing, installation, testing, and operational deployment of such components and systems. Intended as a complementary reference to DNV-ST-F101 – Submarine Pipeline Systems, DNV-RP-F113 aims to ensure that the safety requirements associated with pipeline repair operations are aligned with the standards set forth in DNV-ST-F101. For design and testing of pressure retaining components the DNV-RP-F113 defers to the industry accepted standards. ASME Boiler and Pressure Vessel Code VIII Div. 2 – Rules for construction of pressure vessels is typically referred to.[3]

The RP draws extensively on DNV-RP-A203 – Technology Qualification to define the qualification process applicable to both novel and established technologies. The Recommended Practice DNVGL-RP-A203 – Technology Qualification outlines a structured and systematic methodology for qualifying

components, equipment, and systems that incorporate novel technologies not yet addressed by existing standards. The principal objective of this framework is to enhance confidence in the deployment of innovative solutions by ensuring their reliability and safety through a rigorous qualification process. The core elements of the technology qualification procedure include the definition of the Technology Qualification Basis, a comprehensive Technology Assessment, and the execution of a detailed Technology Qualification Plan. These steps collectively facilitate the identification and mitigation of potential risks associated with new technologies, thereby supporting their responsible integration into operational environments.

The Recommended Practice DNV-CP-0338 – Type Approval Scheme defines the procedural framework through which DNV verifies compliance of standard designs and routinely manufactured, identical components with applicable governing rules. This Type Approval process is applicable to materials, individual products, product families, and systems. As part of the qualification process, a type testing program must be conducted and witnessed to confirm that the material, product, or system meets the specified requirements. Upon successful completion of the evaluation process, a Type Approval Certificate is issued. Subsequently, a DNV Product Certificate may be granted for each manufactured item, contingent upon DNV's involvement in the production process. The certification procedure is detailed in DNV-SE-0499 – Certification of Pipeline Components, which outlines the necessary steps for DNV's participation and oversight during manufacturing activities.

This paper highlights the Saipem Grip & Metal Seal Connector (GMSC) as a representative example of the latest advancements in mechanical connector technology. The GMSC utilizes a metal-to-metal end-seal system that preserves the integrity of the corrosion-resistant alloy (CRA) sealing envelope while maintaining the full internal bore of the pipeline. A comprehensive qualification program for the GMSC—including test rig configuration, applied load cases, instrumentation and monitoring protocols, and acceptance criteria—is detailed in OTC-32193-MS [3]. The referenced study also presents the finite element analysis (FEA) model employed to evaluate the connector's structural performance under simulated operational conditions.

The overall qualification program included a Technology Qualification with DnV comprising a wide number of physical tests on a 12" pipe connector, not reproducing the final GMSC configuration, but specifically designed to allow performing the agreed qualification activities (Figure 9). Additional small-scale tests have also been carried out to verify the actual behavior of some specific components like: gripping teeth, hydraulic wrenches/tensioners used for initial activation, elastomeric seal used for back-seal tests, etc. Practically all physical tests have been reproduced by suitable FEA models in order to improve the modelling skill on the matter and get a better confidence with the intimate behavior of the new connection system.

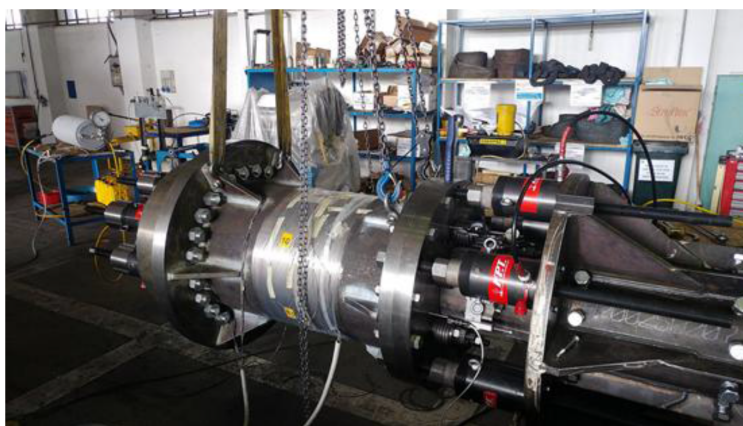


Figure 9—12" pipe connector – test rig

Following completion of the TQ process, certified by DnV and aimed at qualifying the innovative parts of the system, the GMSC underwent further qualification and achieved Type Approval Certification based on detailed design, fabrication and full-scale testing of a connector prototype for a pipe size representative of the identified TA range. A 26" pipeline was selected for developing the detail design activity of the GMSC prototype and for the procurement of the pipe joints required to perform the full-scale test program. The same pipe was used to obtain the spool pieces suitable for carbon steel (with corrosion allowance) and clad pipe tests.

The Saipem Grip & Metal Seal Connector (GMSC) is a subsea pipeline repair connector, remote or diver installed, which is qualified as permanent pipeline repair solution for sour and corrosive service and environment. The GMSC is DNV-GL Type Approved for use as pipeline repair connector for Corrosion Resistant Alloy (CRA) clad- or lined submarine pipelines to DNV-ST-F101, DNV-RP-F113, 2021 ASME Code VIII Div. 2, DNV-RP-B401. Type approval covers from 20 in, to 38 in pipeline sizes, up to 360 bar design pressure, water depth up to 2000 m. The GMSC is designed for sweet and sour service according to ISO 15156-2. The Saipem Grip & Metal Seal Connector (GMSC) is available also for use as pipeline repair connector for carbon steel subsea pipelines with / without corrosion allowance, pipeline sizes from 20 in to 38 in, up to 300 bar design pressure, water depth up to 2000 m.

The MORGRIP CLiP Connector is a qualified as permanent pipeline repair solution DNV-GL Type Approved to OS-F101/ RP-F113, for Pipeline sizes from 10 in to 26 in, maximum design temperature of 149 °C and maximum working pressure of 364 bar. NACE compliant to MR 0175 / ISO-15156.

Conclusions

The Middle East/Arabian Gulf area is currently planning and building a large number of gas development projects, many of which require clad or lined CRA pipe (e.g. Qatar North Field Sustainability, ADNOC Hail & Gasha - the world's largest offshore sour gas development). Most of this infrastructure is critical to local and international energy security requiring robust planning and significant pre-investment in EPRS systems. Furthermore, these projects require relatively large bore long trunklines for which a connector repair is usually more applicable, even in the shallower waters of the Arabian Gulf[1]. All in all, there is an increasing demand for high integrity and sour service rated subsea pipeline connectors as part of the critical repair readiness inventory.

The proven viability of a repair method based on subsea connectors started facing important challenges with the increasing use of sour service pipelines. The first solutions, based on elastomeric seals and largely used in mid-stream repair scenario, were recognized as temporary repair and not suitable for sour service, leading to use of the graphite seal, which overcome the limitation of the elastomeric due to degradation, thus being the first permanent repair method, however despite being compatible from a chemical point of view with corrosive fluid, relies on the external pipe surface for sealing. This does not prevent the carbon steel mother pipe from being in contact with the pipeline fluid and therefore does not fully suit the new pipeline technology for sour service using CRA lined or clad pipe.

The graphite seal greatly improves the resistance of the seal to corrosive fluids, but still permits the pipeline contents to come into direct contact with the carbon steel carrier pipe, which again is not fully suited to CRA lined or clad pipe in sour service.

Recently new repair solutions have been qualified according to the most stringent DnV standards, purposely engineered for CRA clad/lined pipelines. The most prominent solutions rely on frontal seal technology to cope with the internal CRA layer of the sour service pipeline, the two main products on the market are the Clad & Lined Pipe (CLiP) Connector by CSS and the Grip and Metal Seal Connector (GMSC) by Saipem, to note that the first solution present a reduction in internal diameter (ID) while the latter restores the characteristics of the original pipeline even in terms of ID, so maintaining the full pigability of the line. The availability of this technology has allowed a more in-depth assessment of the pros and cons versus

conventional methods, and some of these elements have been addressed in this paper, including comments on safety and availability, applicability as well as implied spare pipe requirements. All in all, it has been shown that introduction of the GMSC can have a significant beneficial impact for the offshore industry, in particular for the development of a comprehensive and reliable EPRS strategy which puts at the top the optimization of applicability and costs.

Nomenclature

CRA	Corrosion Resistant Alloy
CWC	Concrete Weight Coating
DNV	Det Norske Veritas
DNV-GL	Det Norske Veritas – Germanischer Lloyd (renamed as DNV since 2021)
EPRS	Emergency Pipeline Repair System
FEA	Finite Element Analysis
FKM	Fluorine Kautschuk Material (Fluoro-Elastomer)
GMSC	Grip and Metal seal di Saipem per tubi CRA (Saipem Grip & Metal Seal Connector)
HNBR	Hydrogenated Nitril Butadiene Rubber
ID	Internal Diameter
NACE	National Association of Corrosion Engineers
OD	Outside Diameter
NORSOK	Norsk Søkkel Konkurransesepisjon, (Norwegian technical standards pertaining to oil & gas)
TA	Type Approval
TQ	Technical Qualification
WT	Wall Thickness

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